Class 5

Matrix Product States

- DMRG provides $|\psi_{GS}\rangle$ in the following form:

$L = \ell + r + 2$

$$|\psi_{GS}\rangle = \sum_{\alpha_\ell=1}^{M} \sum_{\beta_r=1}^{M} \sum_{\sigma_{\ell+1}\sigma_{\ell+2}=1}^{d} W_{\alpha_\ell \beta_r}^{\sigma_{\ell+1}\sigma_{\ell+2}} \; |\alpha_\ell\rangle \otimes |\sigma_{\ell+1}\rangle \otimes |\sigma_{\ell+2}\rangle \otimes |\beta_r\rangle$$

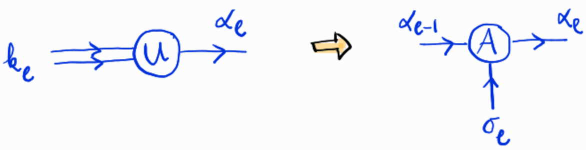
- The basis $|\alpha_\ell\rangle$, $|\beta_r\rangle$ were formed in previous iterations:

→ product basis	$\|k_\ell\rangle = \|\alpha_{\ell-1}\rangle \otimes \|\sigma_\ell\rangle$	$\|j_r\rangle = \|\sigma_{\ell+3}\rangle \otimes \|\beta_{r-1}\rangle$
→ diagonalization of H and partial trace	$\rho^{(\ell)} = \sum_{k_\ell k_\ell'} \rho^{(\ell)}_{k_\ell k_\ell'} \|k_\ell\rangle\langle k_\ell'\|$	$\rho^{(r)} = \sum_{j_r j_r'} \rho^{(r)}_{j_r j_r'} \|j_r\rangle\langle j_r'\|$
→ full diagonalization of $\rho^{(\ell)}$ / $\rho^{(r)}$	$\rho^{(\ell)} = \sum_{\alpha_\ell} W^{(\ell)}_{\alpha_\ell} \|\alpha_\ell\rangle\langle\alpha_\ell\|$ $\|\alpha_\ell\rangle = \sum_{k_\ell} U_{k_\ell \alpha_\ell} \|k_\ell\rangle$	$\rho^{(r)} = \sum_{\beta_r} W^{(r)}_{\beta_r} \|\beta_r\rangle\langle\beta_r\|$ $\|\beta_r\rangle = \sum_{j_r} V^*_{j_r \beta_r} \|j_r\rangle$
→ Truncation ⇓ keeping M dominant eigenvectors U, V are not unitaries but isometries	$\sum_{k_\ell} U^*_{k_\ell \alpha_\ell'} U_{k_\ell \alpha_\ell} = \delta_{\alpha_\ell' \alpha_\ell}$ but $\sum_{\alpha_\ell=1}^{M} U_{k_\ell \alpha_\ell} U^*_{k_\ell' \alpha_\ell} = P^{(\ell)}_{k_\ell k_\ell'}$ projector of kept states	$\sum_{j_r} V_{j_r \beta_r'} V^*_{j_r \beta_r} = \delta_{\beta_r' \beta_r}$ but $\sum_{\beta_r=1}^{M} V^*_{j_r \beta_r} V_{j_r' \beta_r} = P^{(r)}_{j_r j_r'}$

- k_ℓ / j_r are product indices!

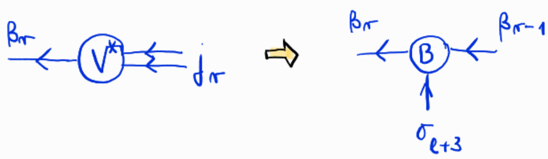
  ↳ reshape U / V to three leg tensors!

$$U_{k_\ell \alpha_\ell} \Longrightarrow A^{(\ell)\,\sigma_\ell}_{\alpha_{\ell-1}\alpha_\ell}$$



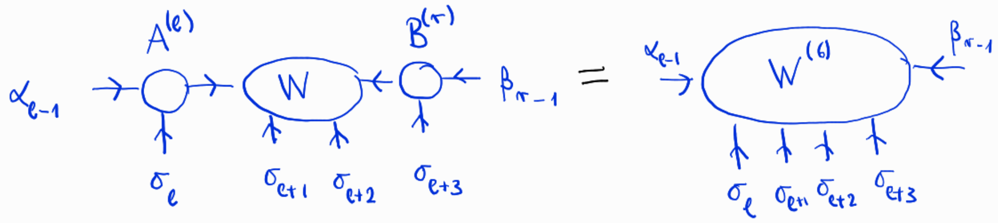
$$V^*_{j_r \beta_r} \Longrightarrow B^{(r)\,\sigma_{\ell+3}}_{\beta_r \beta_{r-1}}$$

here we also permuted the legs!



- Rewrite the Ground state:

$$|\psi_{GS}\rangle = \sum_{\substack{\alpha_{\ell-1}\alpha_\ell \\ \beta_{r-1}\beta_r}} \sum_{\substack{\sigma_\ell\sigma_{\ell+1} \\ \sigma_{\ell+2}\sigma_{\ell+3}}} A^{(\ell)\sigma_\ell}_{\alpha_{\ell-1}\alpha_\ell} W^{\sigma_{\ell+1}\sigma_{\ell+2}}_{\alpha_\ell\beta_r} B^{(r)\sigma_{\ell+3}}_{\beta_r\beta_{r-1}} |\alpha_{\ell-1}\rangle\otimes|\sigma_\ell\rangle\otimes|\sigma_{\ell+1}\rangle\otimes|\sigma_{\ell+2}\rangle\otimes|\sigma_{\ell+3}\rangle\otimes|\beta_{r-1}\rangle$$



- We can iterate this, and rewrite $|\psi_{GS}\rangle$ in the original basis

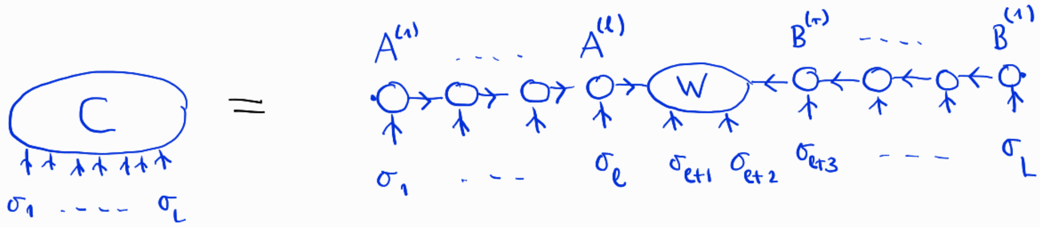
$$|\psi_{GS}\rangle = \sum_{\alpha_1\cdots\alpha_\ell}\sum_{\beta_1\cdots\beta_r}\sum_{\sigma_1\cdots\sigma_L} A^{(1)\sigma_1}_{\cdot\,\alpha_1} A^{(2)\sigma_2}_{\alpha_1\alpha_2}\cdots A^{(\ell)\sigma_\ell}_{\alpha_{\ell-1}\alpha_\ell} W^{\sigma_{\ell+1}\sigma_{\ell+2}}_{\alpha_\ell\beta_r} B^{(r)\sigma_{\ell+3}}_{\beta_r\beta_{r-1}}\cdots B^{(1)\sigma_L}_{\beta_1\,\cdot}$$

$$\cdot\ |\sigma_1\rangle\otimes|\sigma_2\rangle\otimes\cdots\otimes|\sigma_L\rangle$$

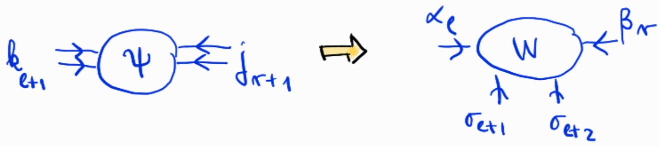
compare this with

$$|\psi_{GS}\rangle = \sum_{\sigma_1\cdots\sigma_L} C_{\sigma_1\cdots\sigma_L} |\sigma_1\rangle\otimes|\sigma_2\rangle\otimes\cdots\otimes|\sigma_L\rangle$$

$\Rightarrow$



this is (almost) an MPS!

- 

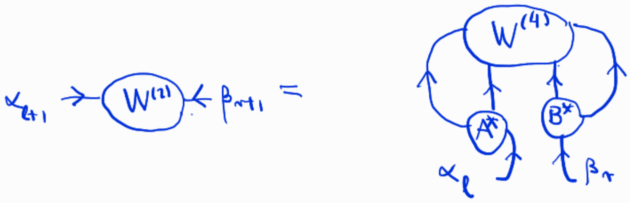
  with $k_{\ell+1} = (\alpha_\ell\,\sigma_{\ell+1})$
  $j_{r+1} = (\sigma_{\ell+2}\,\beta_r)$
  product indices

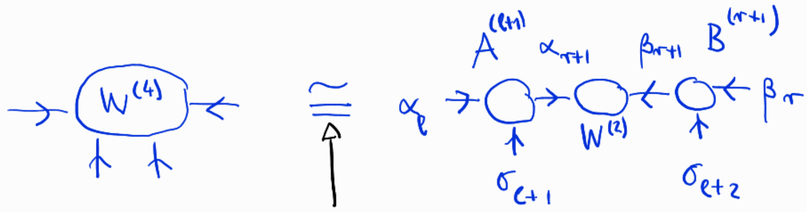
- diagonalize $\rho^{(\ell+1)}$ & $\rho^{(r+1)}$ $\Rightarrow$ $A^{(\ell+1)\sigma_{\ell+1}}_{\alpha_\ell\alpha_{\ell+1}}$
  $B^{(r+1)\sigma_{\ell+2}}_{\beta_{r+1}\beta_r}$

- form the core tensor with no physical legs

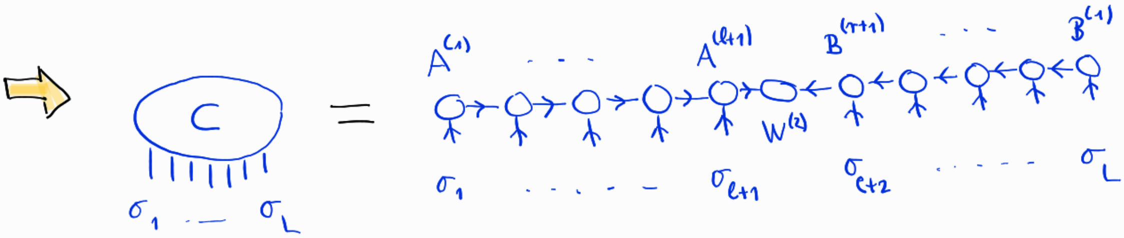
$$W^{(2)}_{\alpha_{\ell+1}\beta_{r+1}} = \sum_{\substack{\alpha_\ell\beta_r \\ \sigma_{\ell+1}\sigma_{\ell+2}}} \left[A^{(\ell+1)\sigma_{\ell+1}}_{\alpha_\ell\alpha_{\ell+1}}\right]^* \left[B^{(r+1)\sigma_{\ell+2}}_{\beta_{r+1}\beta_r}\right]^* W^{(4)\sigma_{\ell+1}\sigma_{\ell+2}}_{\alpha_\ell\beta_r}$$

This is the coefficient tensor of $|\psi_{GS}\rangle$ in the $|\alpha_{\ell+1}\rangle\otimes|\beta_{r+1}\rangle$ basis.



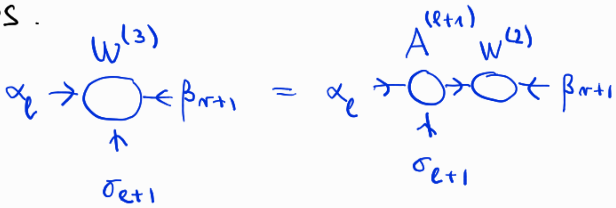


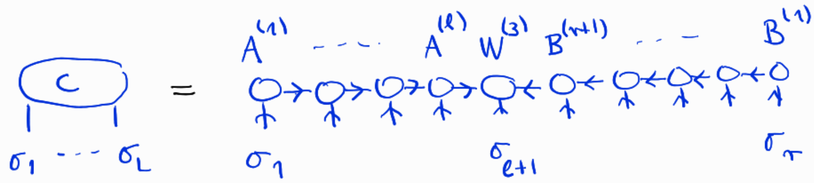
equal,
if no truncation after diagonalization of $\rho^{(\ell+1)}$ & $\rho^{(r+1)}$



- remark: $W^{(2)}$ can be contracted with either $A^{(\ell+1)}$ or $B^{(r+1)}$ to form a 'standard' MPS.

e.g.





origin of name MPS :

$$C_{\sigma_1\sigma_2\ldots\sigma_L} = \underline{\underline{A}}^{(1)\sigma_1}\,\underline{\underline{A}}^{(2)\sigma_2}\ldots\underline{\underline{A}}^{(\ell)\sigma_\ell}\,\underline{\underline{W}}^{(3)\sigma_{\ell+1}}\,\underline{\underline{B}}^{(r+1)\sigma_{\ell+2}}\ldots\underline{\underline{B}}^{(1)\sigma_L}$$

- $M \times M$ size is reached only in the middle (from site n if: $d^n > M$)

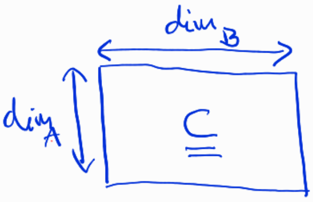
In the following, we turn our viewpoint

- We show that every $C_{\sigma_1\cdots\sigma_L}$ can be expressed as an MPS
- We quantify the errors caused by truncation
- We derive DMRG from the variational principle
- We show, how expectation values and correlators are calculated

The Schmidt decomposition

- Hilbert space formed by two subsystems: $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$

- Arbitrary state vector $|\psi\rangle = \sum_{i=1}^{\dim_A} \sum_{j=1}^{\dim_B} C_{ij} \, |i\rangle \otimes |j\rangle$



<u>Reminder</u> Singular Value Decomposition

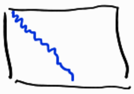
- exists for any matrix irrespective to shape and size

$\underline{\underline{C}} = \underline{\underline{U}} \cdot \underline{\underline{\Lambda}} \cdot \underline{\underline{V}}^{+}$

$\underline{\underline{U}}$: $\dim_A \times \dim_A$ unitary

$\underline{\underline{V}}$: $\dim_B \times \dim_B$ unitary

$\underline{\underline{\Lambda}}$: $\dim_A \times \dim_B$ diagonal



or

- Properties:
  - ⊳ not unique (we will see below)
  - ⊳ diagonal elements of $\underline{\underline{\Lambda}}$ are non-negative and real

$$\underline{\underline{\Lambda}} = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \end{pmatrix}$$

$\lambda_1 \geq \lambda_2 \geq \dots$

and this set is unique

- Plug the SVD in the definition of $|\psi\rangle$

$$|\psi\rangle = \sum_{i=1}^{\dim_A} \sum_{j=1}^{\dim_B} \sum_{\alpha=1}^{\min(\dim_A, \dim_B)} U_{i\alpha} \lambda_\alpha V^*_{j\alpha} \, |i\rangle \otimes |j\rangle =$$

$$= \sum_{\alpha=1}^{\min(\dim_A, \dim_B)} \lambda_\alpha \underbrace{\left(\sum_{i=1}^{\dim_A} U_{i\alpha} |i\rangle \right)}_{\text{def: } |\alpha\rangle_A} \otimes \underbrace{\left(\sum_{j=1}^{\dim_B} V^*_{j\alpha} |j\rangle \right)}_{|\alpha\rangle_B}$$

Schmidt states

⇓

$$\boxed{|\psi\rangle = \sum_{\alpha=1}^{\min(\dim_A, \dim_B)} \lambda_\alpha \, |\alpha\rangle_A \otimes |\alpha\rangle_B}$$

Schmidt-decomposition

- $|\alpha\rangle_A$ & $|\alpha\rangle_B$ form orthonormal basis in $\mathcal{H}_A$ & $\mathcal{H}_B$, respectively

- Connection to reduced density matrices

$$\rho_A = \mathrm{Tr}_B\left(|\psi\rangle\langle\psi|\right) = \mathrm{Tr}_B\left(\sum_\alpha \sum_\beta \lambda_\alpha \lambda_\beta \, |\alpha\rangle_A {}_A\langle\beta| \otimes |\alpha\rangle_B {}_B\langle\beta|\right) = \sum_\alpha \lambda_\alpha^2 \, |\alpha\rangle_A {}_A\langle\alpha|$$

$$\rho_B = \mathrm{Tr}_A\left(|\psi\rangle\langle\psi|\right) = \dots = \sum_\alpha \lambda_\alpha^2 \, |\alpha\rangle_A {}_A\langle\alpha|$$

→ The Schmidt states are eigenstates of the corresponding reduced density matrix

→ The spectra of ρ_A & ρ_B are identical: $w_\alpha = \lambda_\alpha^2$

↳ This is true for any pure state $|\psi\rangle\langle\psi|$!

<u>TRUNCATION</u> keep only M Schmidt pairs

$$|\tilde{\psi}\rangle = \sum_{\alpha=1}^{M} \lambda_\alpha |\alpha\rangle_A \otimes |\alpha\rangle_B$$

<u>Truncation error</u>

$$\boxed{\left\| |\tilde{\psi}\rangle - |\psi\rangle \right\|^2 = \underbrace{\langle\tilde{\psi}|\tilde{\psi}\rangle}_{\sum_{\alpha=1}^{M}\lambda_\alpha^2} + \underbrace{\langle\psi|\psi\rangle}_{\sum_{\alpha=1}^{\text{"}\infty\text{"}}\lambda_\alpha^2} - \underbrace{\langle\psi|\tilde{\psi}\rangle}_{\sum_{\alpha=1}^{M}\lambda_\alpha^2} - \underbrace{\langle\tilde{\psi}|\psi\rangle}_{\sum_{\alpha=1}^{M}\lambda_\alpha^2} = \sum_{\alpha=M+1}^{\text{"}\infty\text{"}} \lambda_\alpha^2}$$

<u>Existence of MPS representation</u>

$$\mathcal{H} = (\mathcal{H}_1)^{\otimes L} = \mathcal{H}_1 \otimes \mathcal{H}_1 \otimes \dots \otimes \mathcal{H}_1$$

$\dim(\mathcal{H}_1) = d$

$\dim(\mathcal{H}) = d^L$

<u>Statement</u>: $\forall \; |\psi\rangle = \sum_{\sigma_1 \dots \sigma_L} C_{\sigma_1 \dots \sigma_L} |\sigma_1\rangle \otimes |\sigma_2\rangle \otimes \dots \otimes |\sigma_L\rangle$

$\exists \; \{ \underline{\underline{M}}^{(1)\sigma_1}, \underline{\underline{M}}^{(2)\sigma_2}, \dots, \underline{\underline{M}}^{(L)\sigma_L} \}$ such, that

$$C_{\sigma_1 \dots \sigma_L} = \underbrace{\underline{\underline{M}}^{(1)\sigma_1}}_{\substack{1\times n \\ n\le d}} \underline{\underline{M}}^{(2)\sigma_2} \dots \underbrace{\underline{\underline{M}}^{(L)\sigma_L}}_{\substack{m\times 1 \\ m\le d}}$$

Proof by construction

- form matrix from C by reshaping : $C_{\sigma_1 \underbrace{(\sigma_2 \cdots \sigma_L)}_{j_{L-1} = 1 \ldots d^{L-1}}} = C_{\sigma_1 j_{L-1}}$

- perform SVD

$$C_{\sigma_1 j_{L-1}} = \sum_{\alpha_1} U^{(1)}_{\sigma_1 \alpha_1} \underbrace{\lambda^{(1)}_{\alpha_1} V^{(1)*}_{j_{L-1} \alpha_1}}_{\Downarrow}$$

- $A^{(1)\sigma_1}_{\cdot \alpha_1} = U^{(1)}_{\sigma_1 \alpha_1}$ $\qquad C^{(1)}_{\alpha_1 \sigma_2 \sigma_3 \cdots \sigma_L} := \lambda^{(1)}_{\alpha_1} V^{(1)*}_{j_{L-1} \alpha_1}$

$j_{L-1} \to \sigma_2 \cdots \sigma_L$ reshaping

• NO SUMMATION FOR α_1 !

- for $\ell = 2 : L-1$
  - → reshape $C^{(\ell-1)}_{\alpha_{\ell-1}\sigma_\ell \cdots \sigma_L}$ as $C^{(\ell-1)}_{\underbrace{(\alpha_{\ell-1}\sigma_\ell)}_{k_\ell}\underbrace{(\sigma_{\ell+1} \cdots \sigma_L)}_{j_{L-\ell}}} = C^{(\ell-1)}_{k_\ell j_{L-\ell}}$
  - → SVD
  $$C^{(\ell-1)}_{k_\ell j_{L-\ell}} = \sum_{\alpha_\ell} U^{(\ell)}_{k_\ell \alpha_\ell} \underbrace{\lambda^{(\ell)}_{\alpha_\ell} V^{(\ell)*}_{j_{L-\ell} \alpha_\ell}}_{\Downarrow}$$
  - → $A^{(\ell)\sigma_\ell}_{\alpha_{\ell-1}\alpha_\ell} := U^{(\ell)}_{k_\ell \alpha_\ell}$ $\qquad C^{(\ell)}_{\alpha_\ell \sigma_{\ell+1} \cdots \sigma_L} := \lambda^{(\ell)}_{\alpha_\ell} V^{(\ell)}_{j_{L-1} \alpha_\ell}$

  $k_\ell \to \alpha_{\ell-1}\sigma_\ell$ reshape $\qquad j_{L-1} \to \sigma_{\ell+1} \cdots \sigma_L$ reshape

  NO SUMMATION FOR α_ℓ !

end for

- $A^{(L)\sigma_L}_{\alpha_{L-1}\cdot} := C^{(L-1)}_{\alpha_{L-1}\sigma_L}$

$$\hookrightarrow \boxed{C_{\sigma_1 \cdots \sigma_L} = \sum_{\alpha_1 \cdots \alpha_{L-1}} A^{(1)\sigma_1}_{\cdot \alpha_1} A^{(2)\sigma_2}_{\alpha_1 \alpha_2} \cdots A^{(L-1)\sigma_{L-1}}_{\alpha_{L-2}\alpha_{L-1}} A^{(L)\sigma_L}_{\alpha_{L-1}\cdot}} \quad !!$$

Remarks

- We did not truncate anywhere. The max. possible matrix bond dimension is $\dim(\alpha_{L/2}) \leq d^{L/2}$
- We iterated from left to right and got A's that are "left" unitaries
$$\sum_{\alpha_{\ell-1}\sigma_\ell} A^{(\ell)\sigma_\ell}_{\alpha_{\ell-1}\alpha_\ell} \left[A^{(\ell)\sigma_\ell}_{\alpha_{\ell-1}\tilde{\alpha}_\ell}\right]^* = \delta_{\alpha_\ell \tilde{\alpha}_\ell}$$

$\rightarrow$ we can do the same by starting from the right too

$$C_{\sigma_1\cdots\sigma_L} = \sum_{\alpha_1\cdots\alpha_{L-1}} B^{(L)\sigma_1}_{\cdot\,\alpha_{L-1}} B^{(L-1)\sigma_2}_{\alpha_{L-1}\alpha_{L-2}} \cdots B^{(1)\sigma_L}_{\alpha_1\,\cdot}$$

$\rightarrow$ we can also go partly from the left, partly from the right

$\Downarrow$

DMRG form $\underline{\underline{A}}^{(1)\sigma_1}\underline{\underline{A}}^{(2)\sigma_2}\cdots\underline{\underline{A}}^{(\ell+1)\sigma_{\ell+1}}\underline{\underline{\Lambda}}^{(\ell+1)}\underline{\underline{B}}^{(r+1)\sigma_{\ell+2}}\cdots\underline{\underline{B}}^{(1)\sigma_L}$

- MPS is not unique! Gauge freedom

$$C^{\sigma_1\cdots\sigma_L} = \underline{\underline{M}}^{(1)\sigma_1}\underline{\underline{M}}^{(2)\sigma_2}\cdots\underline{\underline{M}}^{(L)\sigma_L} = \underbrace{\underline{\underline{M}}^{(1)\sigma_1}\underline{\underline{X}}^{(1)}}_{\underline{\underline{N}}^{(1)\sigma_1}}\underbrace{\underline{\underline{X}}^{(1)^{-1}}\underline{\underline{M}}^{(2)\sigma_2}\underline{\underline{X}}^{(2)}}_{\underline{\underline{N}}^{(2)\sigma_2}}\underline{\underline{X}}^{(2)^{-1}}\cdots\underline{\underline{X}}^{(L-1)}\underbrace{\underline{\underline{X}}^{(L-1)^{-1}}\underline{\underline{M}}^{(L)\sigma_L}}_{\underline{\underline{N}}^{(L)\sigma_L}}$$

$\rightarrow$ Our construction results in left-canonical / right canonical / mixed canonical

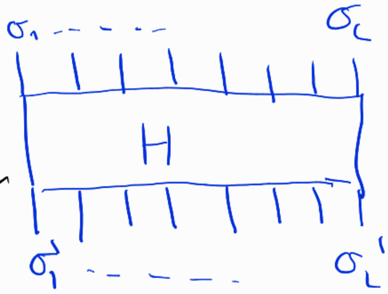
gauge fixing $\Rightarrow$ for non-degenerate Schmidt spectra only phase freedom remains

$\Rightarrow$ rotational freedom in degenerate subspaces

Derivation of 1-site DMRG from variational principle

$$|\psi\rangle = \underline{\underline{M}}^{(1)\sigma_1}\underline{\underline{M}}^{(2)\sigma_2}\cdots\underline{\underline{M}}^{(L)\sigma_L}$$

$$H = \sum_{\substack{\sigma_1\cdots\sigma_L\\ \sigma_1'\cdots\sigma_L'}} H^{\sigma_1'\cdots\sigma_L'}_{\sigma_1\cdots\sigma_L} |\sigma_1'\cdots\sigma_L'\rangle\langle\sigma_1\cdots\sigma_L|$$



$\rightarrow$ never stored in this form!

example

$$H = J\sum_i \sigma_i^x\sigma_{i+1}^x - h\sum_i \sigma_i^z$$

$\rightarrow$ sum of $2L$ simple terms

$$E_{GS} \approx \min_{\{\underline{\underline{M}}\}} \langle\psi|H|\psi\rangle$$

constraint $\langle\psi|\psi\rangle = 1$

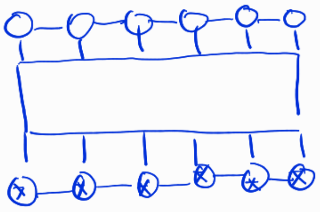
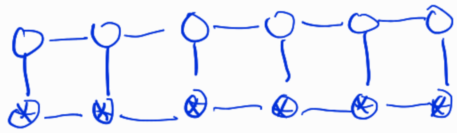
$\Downarrow$

Lagrange multiplier

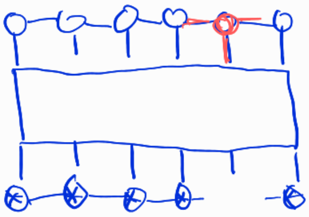
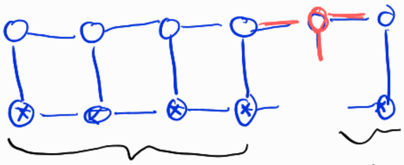
$$\min_{\{\underline{\underline{M}}\}} \underbrace{\left(\langle\psi|H|\psi\rangle - \mu\langle\psi|\psi\rangle\right)}$$

$$\sum_{\substack{\sigma_1\cdots\sigma_L\\ \sigma_1'\cdots\sigma_L'}} \underline{\underline{M}}^{(1)\sigma_1'^*}\cdots\underline{\underline{M}}^{(L)\sigma_L'^*} H^{\sigma_1'\cdots\sigma_L'}_{\sigma_1\cdots\sigma_L}\underline{\underline{M}}^{(1)\sigma_1}\cdots\underline{\underline{M}}^{(L)\sigma_L} -$$

$$-\mu\sum_{\sigma_1\cdots\sigma_L}\underline{\underline{M}}^{(1)\sigma_1^*}\cdots\underline{\underline{M}}^{(L)\sigma_L^*}\underline{\underline{M}}^{(1)\sigma_1}\cdots\underline{\underline{M}}^{(L)\sigma_L}$$

 $-\mu$ 

- iterative optimization of tensors

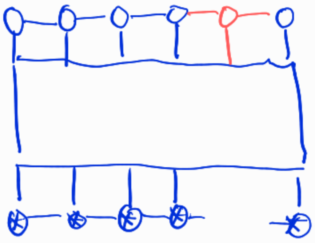
$\frac{\partial}{\partial \underline{\underline{M}}^{(\ell)\sigma_\ell^*}}$: $0 =$  $-\mu$ 

This is a generalized eigen-value problem

is 1 if left canonical

is 1 if right canonical

↳ if we use left canonical tensors left from the optimized one, and right canonical in the right, we get a standard eigenvalue problem

 $-\mu$  $= 0$

- This is 1-site DMRG.

- 2-site DMRG can be derived by optimizing 

  ↳ we leave the MPS manifold at every iteration

  ↳ but come back by truncation

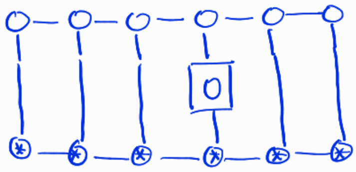
    ↳ we can increase the bond dimension, if we observe large truncation error

    ↳ 2-site DMRG has superior convergence speed compared to 1-site variant.

    ↳ Computational cost is higher ($\dim \mathcal{H}_{eff}^{2site} = d \cdot \dim \mathcal{H}_{eff}^{1site}$)

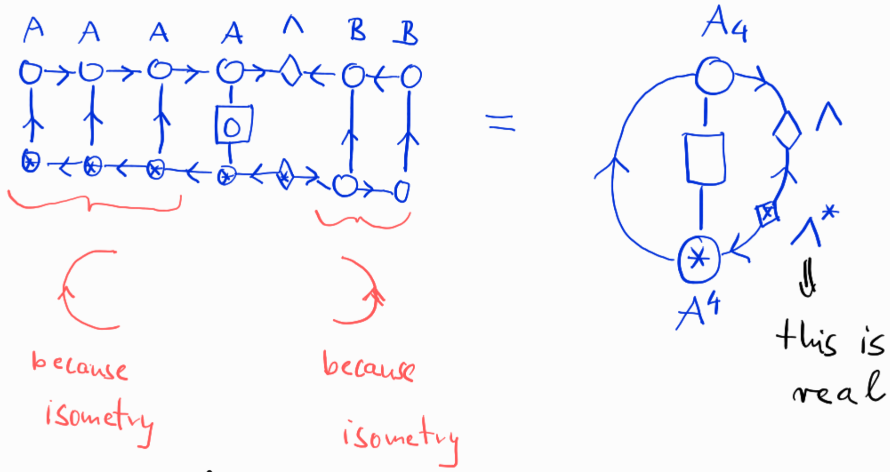
Expectation values:

- 1-site op:

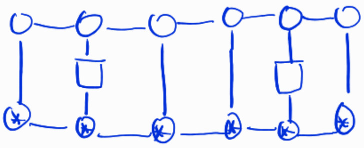


$\langle\psi|O_4|\psi\rangle$

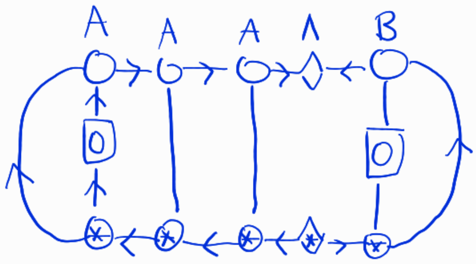
best gauge fixing: left canonical in the left, right canonical in the right



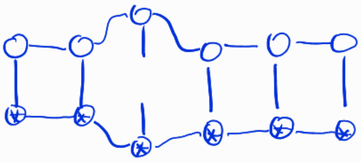
- 2-site correlators are similar

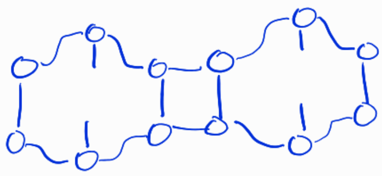


is again simplified in mixed canonical case; e.g.



- 1-site and 2-site RDM's

$\rho^{1site}_3 =$ 

$\rho^{2site}_{2,5} =$ 

- 1-site & 2-site entropies

$$S_3 = -\mathrm{tr}\left(\rho^{1site}_3 \log \rho^{1site}_3\right) \qquad S_{2,5} = -\mathrm{tr}\left(\rho^{2site}_{2,5} \log \rho^{2site}_{2,5}\right)$$

- 2-site mutual information $\boxed{I_{ij} = S_i + S_j - S_{ij}}$ ~ correlation strength between sites i & j